

The Euclidean Algorithm

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1 The program

```
1 unsigned int gcd(unsigned int a, unsigned int b)
2 {
3     while (b != 0) {
4         unsigned int t = a % b;
5         a = b;
6         b = t;
7     }
8     return a;
9 }
```

Throughout, $\text{gcd}(x, y)$ denotes the largest common divisor of x and y , with the conventions $\text{gcd}(x, 0) = x$ and $\text{gcd}(0, 0) = 0$. We write $D(x, y)$ for the set of common divisors of x and y . The arguments supplied by the caller are denoted a_0 and b_0 .

2 Correctness

Lemma 1 (division step). *For all integers $x \geq 0$ and $y > 0$,*

$$D(x, y) = D(y, x \bmod y), \quad \text{and consequently} \quad \text{gcd}(x, y) = \text{gcd}(y, x \bmod y).$$

Proof. Put $q = \lfloor x/y \rfloor$ and $r = x - qy = x \bmod y$.

If $d \mid x$ and $d \mid y$, then d divides $x - qy = r$, so $d \in D(y, r)$. Conversely, if $d \mid y$ and $d \mid r$, then d divides $qy + r = x$, so $d \in D(x, y)$. The two sets therefore coincide. Both are nonempty (they contain 1) and bounded above by $y > 0$, so each has a largest element, and these largest elements are equal. $\square$

Hence, it follows that at every evaluation of the loop condition,

$$\text{gcd}(a, b) = \text{gcd}(a_0, b_0).$$

Termination. We want to prove that the algorithm terminates.

Let b_k denote the value of **b** at the k -th evaluation of the loop condition. Whenever the body executes we have $b_k > 0$, and therefore

$$b_{k+1} = a_k \bmod b_k < b_k.$$

Thus (b_k) is a strictly decreasing sequence of non-negative integers. No such sequence is infinite, so the loop is executed only finitely often.

When the loop exits, the loop condition fails, so $b = 0$, and the invariant reads

$$\text{gcd}(a, 0) = a = \text{gcd}(a_0, b_0).$$

The function returns a , which is the required value. $\square$

3 The step bound

Define the remainder sequence

$$r_0 = a_0, \quad r_1 = b_0, \quad r_{i+1} = r_{i-1} \bmod r_i \quad (\text{defined as long as } r_i > 0).$$

A straightforward induction shows that after k iterations of the loop the variables hold $(a, b) = (r_k, r_{k+1})$. Hence the body executes exactly at those indices $j \geq 1$ for which $r_j \neq 0$, and the total number of iterations is

$$T = m - 1, \quad m = \min\{i \geq 1 : r_i = 0\}.$$

Note that $r_{i+1} = r_{i-1} \bmod r_i < r_i$ for every $i \geq 1$, so the sequence is strictly decreasing *from index 1 onward*. Index 0 must be excluded: if $a_0 < b_0$ then $r_1 = b_0 > r_0$, and the first iteration merely exchanges the two arguments.

Lemma 2. *For every $i \geq 1$ with $r_{i+1} > 0$ we have $r_{i+2} < r_i/2$.*

Proof. From definition we already have $r_{i+1} < r_i$ for $i \geq 1$. Now, we have two cases.

Case 1: $2r_{i+1} \leq r_i$. Then $r_{i+2} = r_i \bmod r_{i+1} < r_{i+1} \leq r_i/2$.

Case 2: $2r_{i+1} > r_i$. We have $1 \leq r_i/r_{i+1} < 2$, so $\lfloor r_i/r_{i+1} \rfloor = 1$ and

$$r_{i+2} = r_i \bmod r_{i+1} = r_i - r_{i+1} < r_i - \frac{1}{2}r_i = \frac{1}{2}r_i. \quad \square$$

Theorem 1. *If a_0 and b_0 are positive n -bit integers, the loop performs at most $2n$ iterations.*

Proof. Since b_0 is an n -bit integer, $r_1 = b_0 \leq 2^n - 1 < 2^n$. Applying Lemma 2 repeatedly, induction on k gives: for every $k \geq 0$ for which r_{1+2k} is defined,

$$r_{1+2k} < \frac{r_1}{2^k} < 2^{n-k}.$$

Take $k = n$. Either the sequence reached 0 at some index at most $2n$, or r_{1+2n} is defined and satisfies $r_{1+2n} < 2^0 = 1$, which forces $r_{1+2n} = 0$. In either case $m \leq 2n + 1$, and therefore

$$T = m - 1 \leq 2n. \quad \square$$